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Human Human Interaction Lab

Filtering the crowd so every hello counts: a reciprocal, proximity aware name badge on the phone you already own

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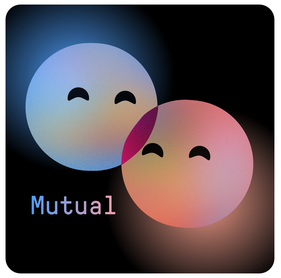
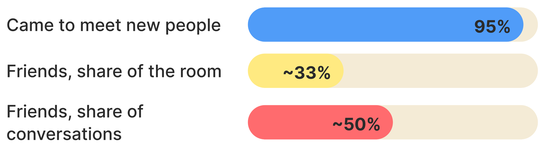


1 The problem

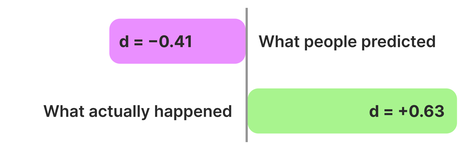
Hackathons, conferences, career fairs, orientation: you get a few hours to find a team, a cofounder or a mentor among hundreds of strangers. Name tags say nothing about fit, so every conversation is a gamble with your limited time, and the right person may be ten feet away.

People don't mix at mixers



Ingram and Morris, Administrative Science Quarterly, 2007. ~100 executives wearing electronic name tags at a mixer.

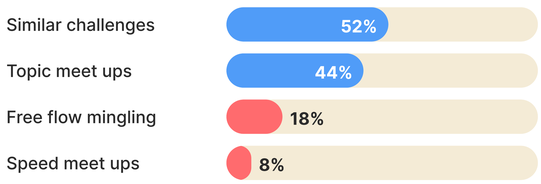
We expect strangers to be worse company than they are



Epley and Schroeder, J. Exp. Psychology: General, 2014. Also: the liking gap, partners like us more than we think (5.82 vs 5.17 of 7; Boothby et al., Psych. Science 2018).

68% of event attendees call meeting new contacts a top networking goal

But they want to be introduced, not left to mingle. What helps attendees network:



Freeman Trends Report 2024, Attendee Intent and Behavior: about 2,100 event attendees, Nov 2023, ±2 pts. Industry survey.

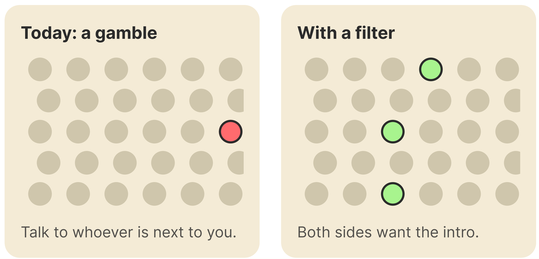
2 Our thinking

THE QUESTION

Can we make each encounter more valuable on average?

Not more conversations. Better odds for each one.

Filtering beats picking at random



Evidence: meeting the right people is the bottleneck. Researchers randomly placed in one shared 90 minute session were 75% more likely to apply for a grant together (Boudreau et al., Review of Economics and Statistics 2017). When a hurricane cancelled APSA 2012, would be attendees became 16% less likely to coauthor (Campos et al., Economic Journal 2018).

What we believe

- **Filter before you talk**
  Reciprocal: an intro only happens when both people would want it.
- **Keep eyes up**
  The signal is worn on your chest, not buried in another feed.
- **Lower the fear**
  A shared color says: the other person wants this too.
- **Close the loop**
  A summary and a follow up arrive by text after you talk.
- **No new hardware**
  The phone you own, in a 3D printed lanyard holder.

Placeholder
Photo or time lapse still: the hackathon crowd

3 Related work

Meme Tags
Borovoy et al., published at CSCW 1998
Did: conference name tags that swapped short ideas face to face.
Inspired: a badge should give strangers something to talk about.
Different: our topics flip to what this person wants to discuss with you.

Our drawing of a smart badge

Social Serendipity
Eagle and Pentland, published in IEEE Pervasive Computing 2005
Did: Bluetooth phones alerted two nearby people whose profiles matched.
Inspired: an intro needs both sides to clear the bar.
Different: an LLM scores needs in plain language, and the cue is worn, not a phone alert.

Figure from Eagle and Pentland, IEEE Pervasive Computing 2005

SocialStools
Guo, Leshed and Green, published at CHI 2023
Did: stools whose distance drives shared sound and light between strangers.
Inspired: map physical distance to a shared cue.
Different: our cue is a color both badges take on, only for mutual matches.

Our drawing of distance as a shared cue

RECON
Pizzato et al., published at RecSys 2010
Did: a reciprocal recommender for online dating.
Inspired: score each direction of interest.
Different: a geometric mean of LLM fit, used in the physical room.

What's new here

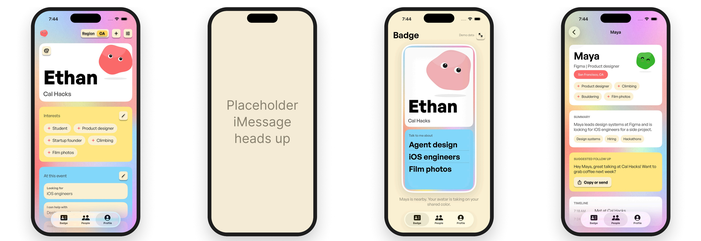
- Matching that needs both sides, scored by an LLM from natural language
- A physical, ambient signal you wear, not a phone alert
- Peer to peer UWB: no venue readers, no custom badge
- The loop closes after the chat: summary, follow up, a personal CRM

Next: measure intros per hour, and how many saved contacts become a real follow up.

4 Our solution

A live badge you wear on a lanyard

How it works

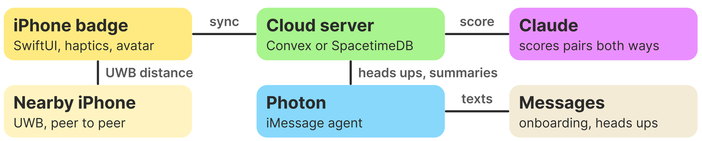


RECIPROCAL SCORE, COMPUTED BY CLAUDE FOR EVERY PAIR

$$\text{score} = 0.75\sqrt{\text{fit}_{\text{me}} \cdot \text{fit}_{\text{them}}} + 0.25 \text{ values}$$

A geometric mean punishes one sided interest: fits of 0.9 and 0.1 score 0.30, below 0.5 and 0.5 at 0.50.

Architecture



References: Ingram & Morris, Administrative Science Quarterly 2007 · Freeman Trends Report 2024 · Epley & Schroeder, JEP: General 2014 · Boothby et al., Psychological Science 2018 · Boudreau et al., Review of Economics and Statistics 2017 · Campos, Leon & McQuillin, Economic Journal 2018 · Borovoy et al., CSCW 1998 · Eagle & Pentland, IEEE Pervasive Computing 2005 · Guo, Leshed & Green, CHI 2023 · Pizzato et al., RecSys 2010

Built at Cal Hacks 13.0 with Convex, Claude, Photon and Figma.
Demo and Devpost: devpost.com/software/human-human-interaction-lab

QR Devpost